

Viral replication and Lab diagnosis

College of medicine / Microbiology unite

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Viruses:

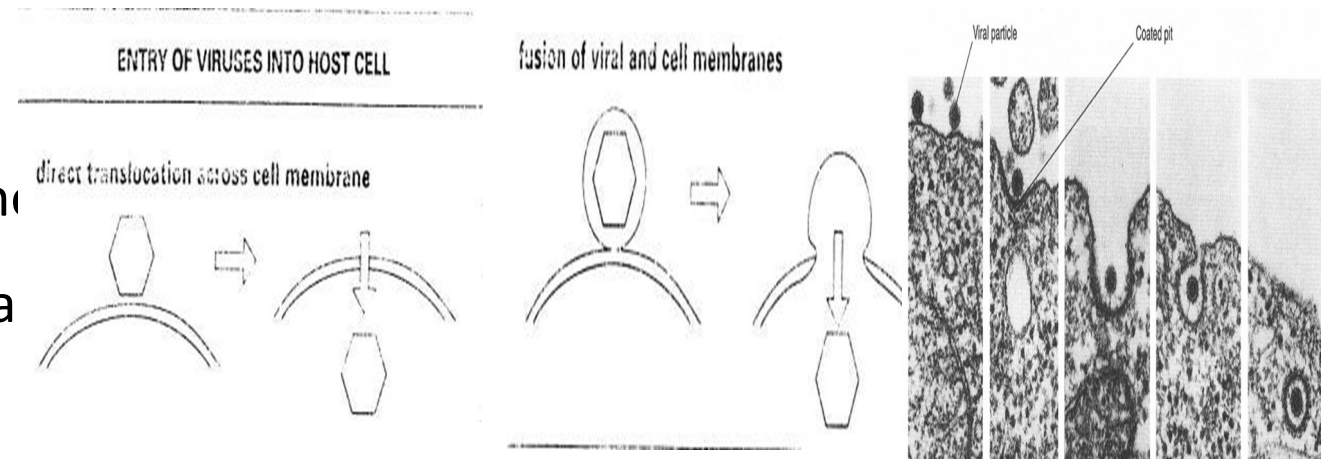
- **Obligate** intracellular organisms with **various tropism**.
- They can be **DNA** or **RNA** genome with a protein capsid and also include cell membrane from the host when they are escaping from.
- Takes over the host **machinery** system for **replication**.

General Steps in Viral Replication Cycles

- **1. Attachment** : The virus must be able to attach to its host and enter the “correct” or “target cell”. A virus exhibit a **tropism** for a particular cell type when it targets and infects that cell type. In many cases these cell types are a specific population of cells within organs.
- **Host range**: refers to the different types of tissue culture cells or organisms (species) that the virus can infect. The host range may be **broad** (infecting many different animals or cell lines of different species) or **narrow**. An example of a broad-range virus is rabies, for which all mammals have varying susceptibility.

- **2. Penetration:**

- a) Direct translocation across cell membrane
- b) Fusion of viral envelope and cell membrane
- c) Engulfed in a pinocytotic vesicle.



3. Uncoating: is the physical separation of the viral nucleic acid from the outer structural components of the virion so that it can function.

4- Gene expression & gene replication : The first step in viral gene expression is mRNA synthesis . At this point viruses follow different pathways:

- **DNA viruses:** replicate in the nucleus and use the host cell DNA – dependent RNA polymerase to synthesize their mRNA . With one exception , The **poxviruses** are the exception because they replicate in the cytoplasm , where they don't have access to the host cell RNA polymerase . They therefore **carry their own polymerase within the virus particle .**

- Most **RNA viruses** undergo their replicative cycle in the cytoplasm . The two exceptions **are retroviruses** and **influenza viruses** , both of which replicate in the **nucleus** .

Strategies for synthesizing mRNA :

- 1) The simplest strategy occurred in virus has a **+ssRNA** (single stranded RNA of positive polarity) positive polarity is defined as an RNA with the same base sequence as the mRNA .
- 2) The second group has **-ssRNA** (single stranded RNA of negative polarity) as its genetic material . Because the cell doesn't have an RNA polymerase capable of using RNA as a template , the virus carries its own **RNA dependent RNA polymerase**.
- 3) The third group has double strand RNA(**dsRNA**) as its genetic material .Because the cell has no enzyme capable of transcribing this RNA into mRNA , the virus carries its own polymerase .(Reovirus).
- 4) The fourth group (retroviruses) has **+ssRNA** that is transcribed into dsDNA by **RNA dependent DNA polymerase (reverse transcriptase)** carried by the virus .

5- Early viral proteins synthesis (Translation): Once the viral mRNA of either DNA or RNA viruses is synthesized, it is translated by host cell ribosomes into viral proteins. Early proteins: Late proteins.

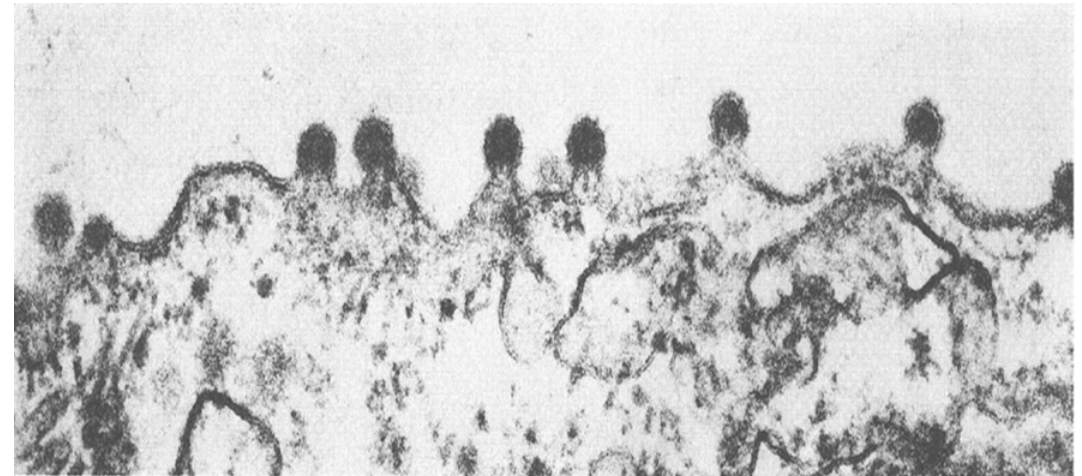
6- Assembly and release:

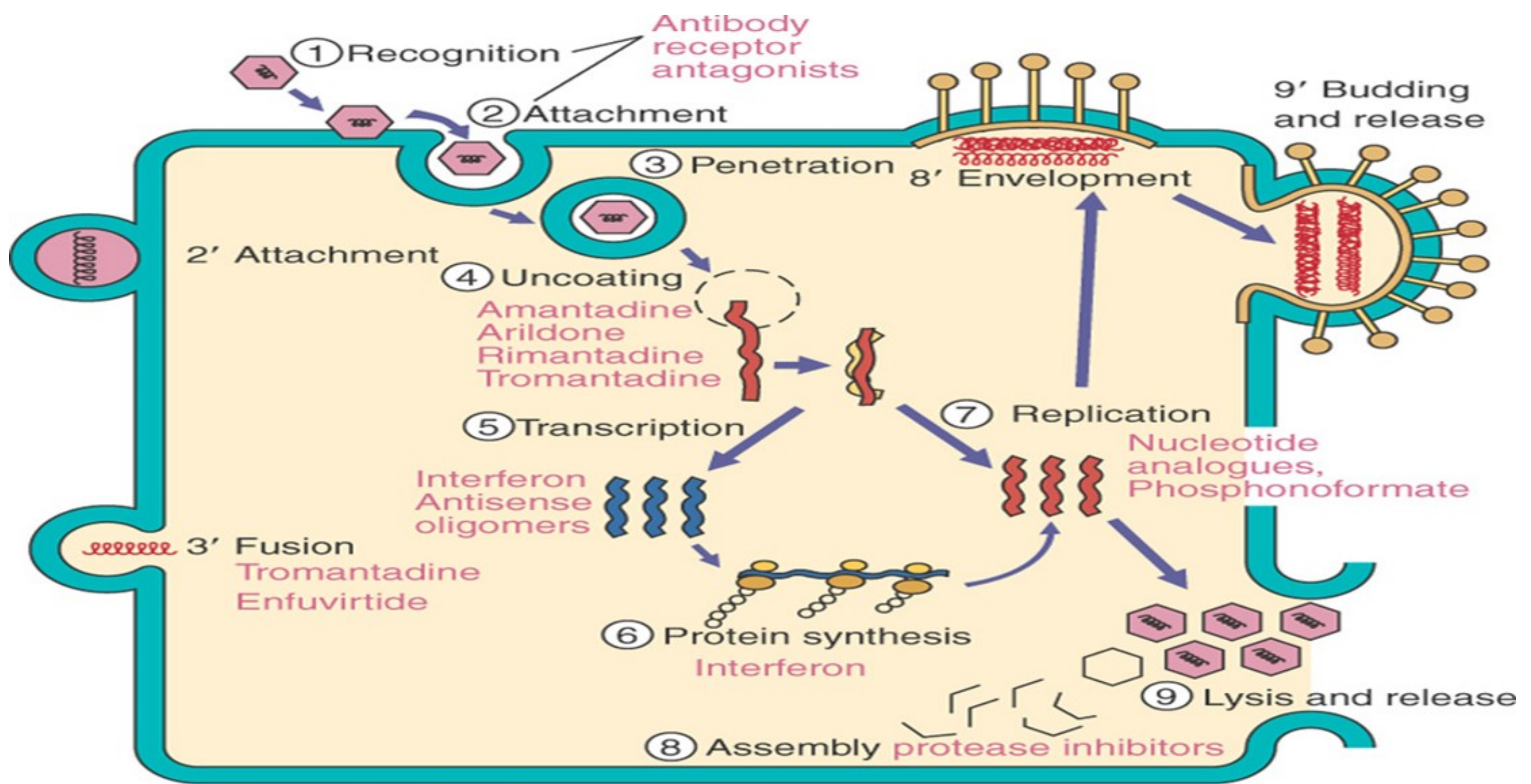
The progeny particles are assembled by packaging the viral nucleic acid within the capsid proteins.

Virus particles are released from the cell by either of two processes:

Rupture of the cell membrane and release of the mature particles (unenveloped viruses).

Budding through the outer cell membrane (enveloped viruses).





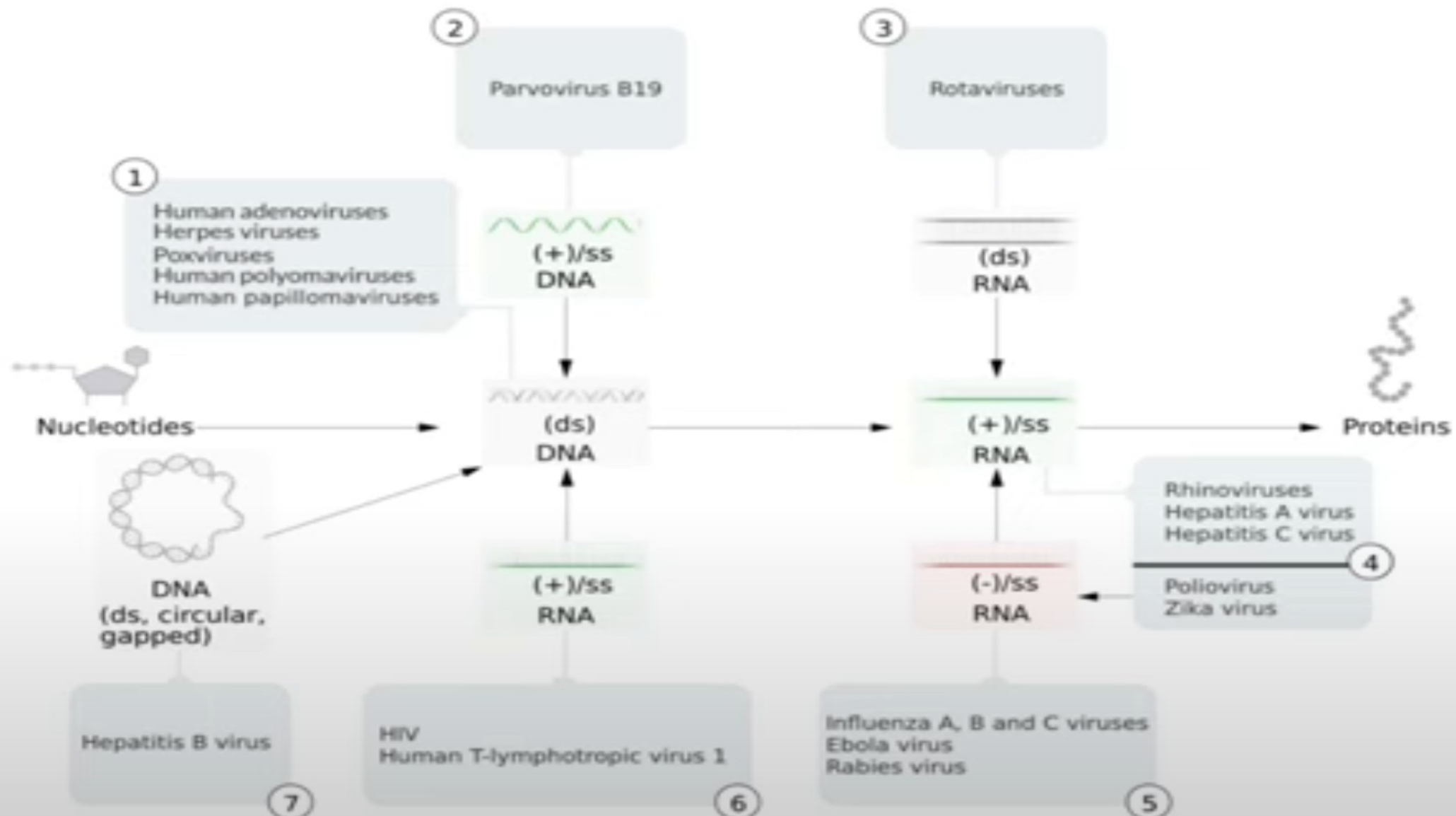
Other major targets:

Nucleotide biosynthesis and mutation: ribavirin

Thymidine kinase (drug activation): acyclovir, penciclovir

Neuraminidase: zanamivir, oseltamivir

Baltimore Classification



Infection

1. **Lytic:**

As the cell hosting the virus; the virus consume all of the host ATP machinery & all of the tropic capacity until the cell dies & lysis.

2. **Persistent:**

the virus persist continually in the cell is not cleared but remains in specific cells (e.g. Hepatitis B).

3. **Latent infection:**

The virus sit in a latent form or circular form or in the form they can't be transcribed.

Pathology

1) Chronic inflammatory infiltration:

- It's intracellular pathogen Abs, neutrophils and macrophages in most cases would not do much to the virus. In this case we need chronic inflammatory cell infiltration, cytotoxic lymphocytes & Nk cells.

2) Cell lysis, cell fusion & inclusion bodies:

- **Cell lysis:** directly can see effects such blisters after epidermal viral infection e.g. herpes varicella virus (chicken pox).
- **Cell fusion:** the virus lysis the membrane between two different cells and appears as multinucleated cell. (**Tzank prep**.: in herpes varicella virus infection, develop blisters during microscope examination shows multinucleated cell it means +ve Tzank prep.

- **Inclusion bodies**: accumulation of viral proteins or viral genomic bodies. You can see collection of eosinophilic (pink) or blue materials within the cell, it means that cell is infected.

Laboratory diagnosis of viral infection:

- 1) Direct detection of viral components either in cells derived from infected tissues or free in specimens.
- 2) Isolation of viruses, usually in cell cultures followed by identification
- 3) Demonstration of the significant increases in serum levels of antibodies to the aetiological viruses during the course of illness.

Criteria for viral specimen diagnosis:

- Timing
- Choice of specimens
- Clinical data
- Specimen transport & storage.

Common Pathogenic Viruses	Specimens for Culture	Comments
Respiratory Tract		
Adenovirus; influenza virus; enterovirus (picornavirus); rhinovirus; paramyxovirus; rubella virus; HSV	Nasal washing, throat swab, nasal swab, sputum	Enterovirus is also shed in stool
Gastrointestinal Tract		
Reovirus; rotavirus; adenovirus; Norwalk virus, calicivirus	Stool, rectal swab	Samples are analyzed by electron microscopy and antigen detection (ELISA); viruses are not cultured
Maculopapular Rash		
Adenovirus; enterovirus (picornavirus)	Throat swab, rectal swab	—
Rubella virus; measles virus	Urine	—
Vesicular Rash		
Coxsackievirus; echovirus; HSV; VZV	Vesicle fluid, scraping, or swab, enterovirus in stool	Initial diagnosis of HSV and VZV can be obtained from vesicle scraping (Tzanck smear)
Central Nervous System (Aseptic Meningitis, Encephalitis)		
Enterovirus (picornavirus)	Stool	PCR
Arboviruses (e.g., togaviruses, bunyavirus)	Rarely cultured	Diagnosis is by serologic tests
Rabies virus	Tissue, saliva, brain biopsy	Diagnosis is by immunofluorescence analysis for antigen
HSV; CMV; mumps virus; measles virus	Cerebrospinal fluid	PCR, virus isolation, and antigen are assayed
Urinary Tract		
Adenovirus; CMV	Urine	CMV may be shed without apparent disease
Blood		
HIV; human T-cell leukemia virus; hepatitis B, C, and D viruses	Blood	Serologic antigen or antibody detection (ELISA), PCR, and RT-PCR are performed

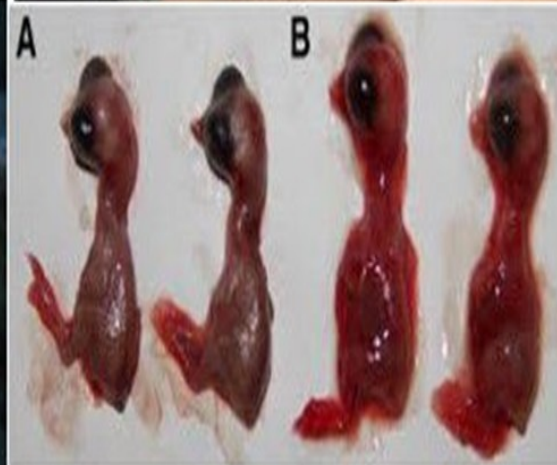
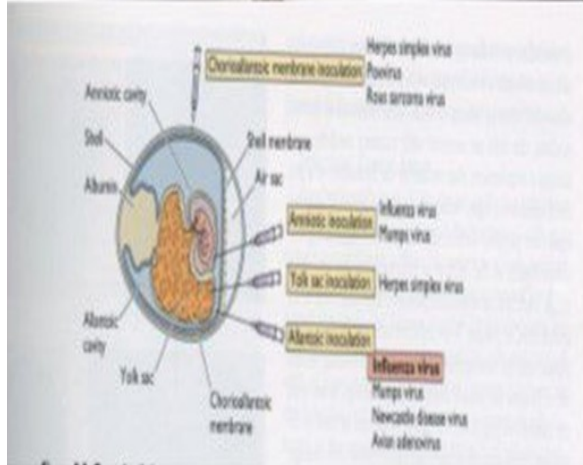
Culture & isolation

➤ The gold standard for viral replication

Viral culture: the viruses do not grow on the artificial medium; they must be cultured on the cell cultures.

1) Embryonated eggs:

- Inoculation of embryonated eggs used for growing of some viruses (influenza virus).
- They examined for the evidence of viral replication: death, pocks(lesion) formation
exanimated of the embryonated egg fluid for the viral antigens.



2) **Laboratory animals:** mice & chimpanzees were used in past but now replaced with cell culture.

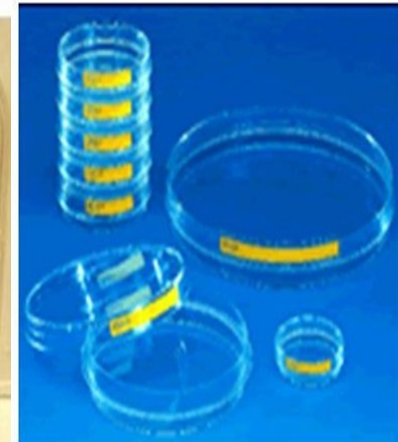
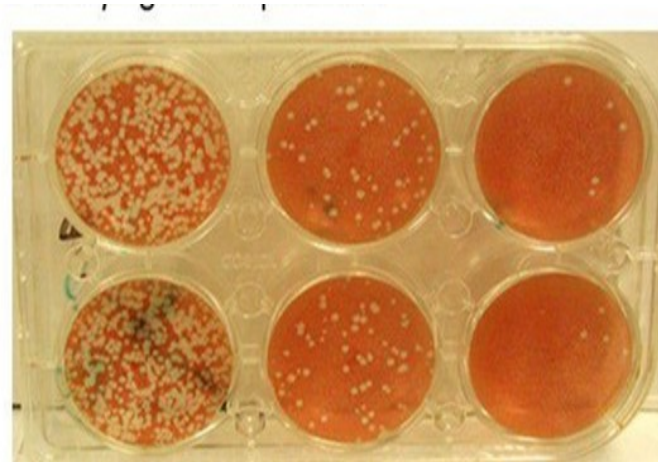
3) **Cell cultures:**

- **Cell culture originated from available host cells.**

- Cells are Kept moist & supplied with nutrients.
- Cell culture incubated for 1-4 weeks, depending on the type or the virus.
- Periodically, they are inspected for the presence of the virus, indicated by areas of dead or dying cells called **cytopathic effect (CPE)**.

- **Cell culture used for:**

- Isolation & growth of the variety of cell
- Studying viral replication.

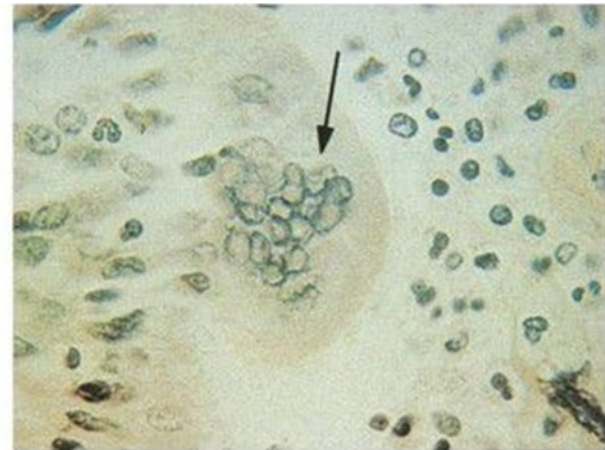
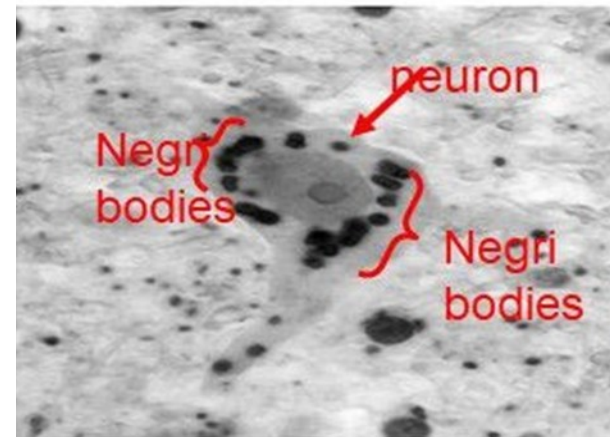


Types of cell cultures:

- 1) **Primary cultures:** made from freshly removed host tissues, they are unable to grow for more than few passages in culture (kidney & lung).
- 2) **Diploid cell lines:** that allows their limited culture (up to 50 passages, embryo).
- 3) **Continuous cell lines:** are cultures for more prolonged (malignant tissues) e.g. Hela cells.

Viral identification in cell cultures:

- 1) Cultured cells changes their morphology & lysis.
- 2) Cytopathic effects of the viruses. (CPE).
- 3) Viruses have distinct CPEs, just as a colonies as bacteria on agar plates have unique morphologies, (after viral cell culturing & examining under microscope for lesions like CPEs or inclusion bodies).
 - Negri bodies are inclusion bodies in case of infection with Rabies.



Rapid diagnostic tests

1) **Direct electron microscopic examination for specimens**, e.g. examination of the stool for Rotavirus & Hepatitis A. (it is not practical for the usual daily clinical works).

2) **Antigen detecting methods:**

Viral antigen detecting by immunofluorescent technique (Ag & Ab reaction).

3) **Agglutination Method:**

Antibodies will be added to the specimens to detect the presence of the viral antigen.

4) **Molecular technique: (PCR)**

Use to detect the viral genome in the clinical specimens.

5) Serology:

- The diagnosis of viral infection by demonstration of antibodies or viral antigens is known as serology. Ways of viral detection via serology:

- Measuring specific viral IgM.
- Measuring specific viral IgG.
- Detection of viral antigens.

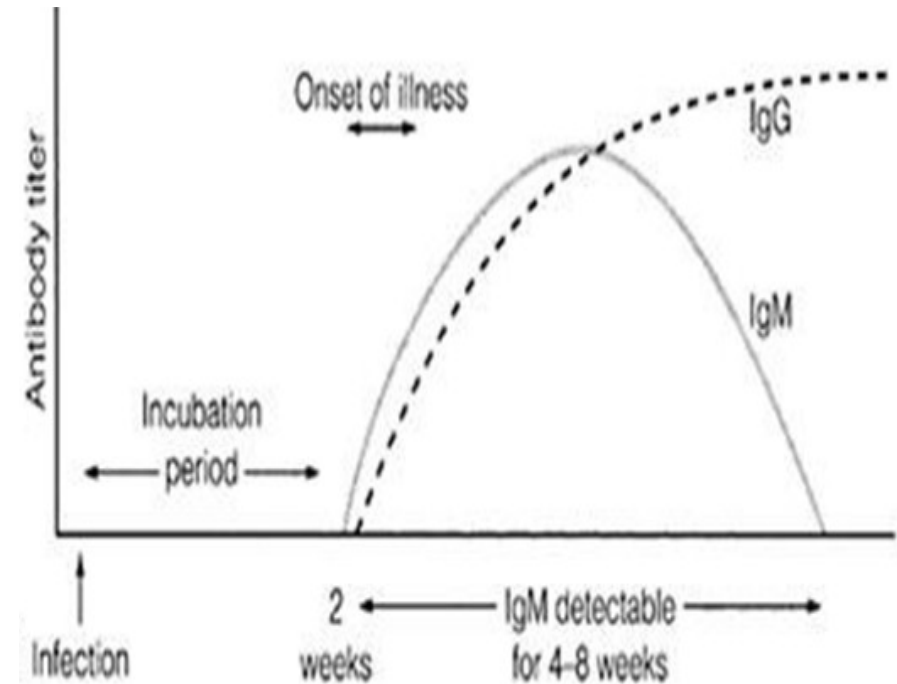
1) Radioimmunoassay (RIA)

2) Enzyme linked Immunosorbent Assay (ELISA) or EIA.

3) indirect immunofluorescent tests.

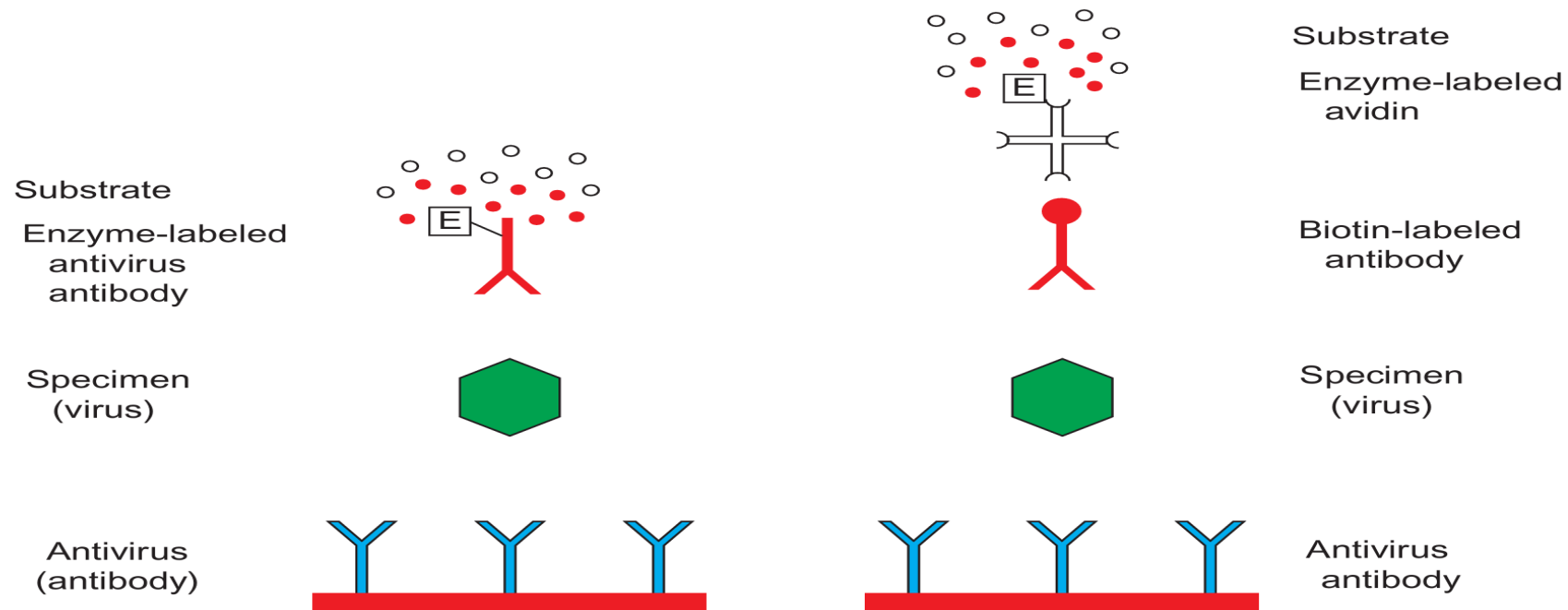
4) Haem agglutination test (Latex agglutination test).

5) Complement fixation test.



Enzyme linked Immunosorbent Assay (ELISA) or EIA.

Laboratory Diagnosis of Virus Diseases



Enzyme immunoassays (EIA or ELISA) for detection of virus and/or viral antigen. *Left:* Direct method. *Right:* Indirect method, using biotinylated antiviral antibody, followed by enzyme- (e.g., peroxidase)- labeled avidin. In each case an enzyme substrate is then added to develop a color reaction. Note the immobilization of the capture antibody on a solid support to facilitate subsequent washing steps. *Reproduced from MacLachlan, N.J., Dubovi, E.J., 2011. Veterinary Virology, fourth ed., Academic Press, with permission.*